

MA 5104: Statistical Inference and Learning

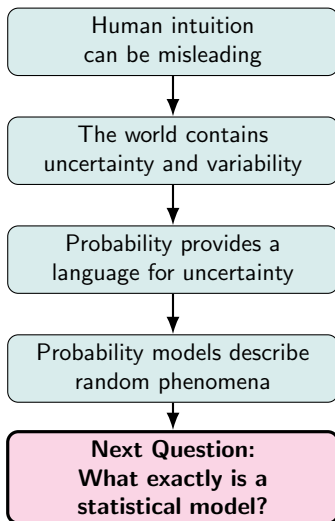
Module 1: Statistical Models and Likelihood

Lecture 3: Understanding Statistical Models

Dr. Rakhi Singh
Department of Mathematics
IIT Madras

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Week 1 Summary



Key Takeaways

By the end of this lecture, you should be able to

- explain what a statistical model is and why statisticians use models;
- interpret the mathematical formulation of a statistical model;
- distinguish observations, random variables, parameters, and modelling assumptions;
- recognize different classes of statistical models, including parametric, semiparametric, nonparametric, generative, and latent variable models;
- appreciate that statistical models are simplified mathematical representations of reality.

Example 1: WYSIATI (What You See Is All There Is)

A company reports

- Revenue growth: 40% per year
- Customer base doubled last year
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- Competition, Data quality
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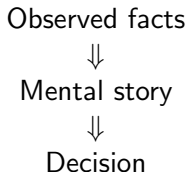
WYSIATI: People construct coherent stories from available information and systematically ignore information that is missing.

Statistics Makes Assumptions Explicit

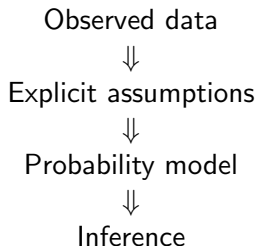
Human reasoning often relies on implicit assumptions.

Statistics requires every important assumption to be stated explicitly.

Informal Reasoning



Statistical Reasoning



Central Question

How do statisticians represent assumptions mathematically?

Representing Assumptions Mathematically

Suppose we observe the systolic blood pressures of n individuals,

$$X_1, \dots, X_n.$$

A statistician proposes the following mathematical description of the data-generating process:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2).$$

Question

What information is encoded in this single mathematical statement?

Decoding the Model

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$$

Notation	Interpretation
$X_1, \dots, X_n$	Random variables representing future observations.
iid	Independent observations from the same (identical) population.
$N(\mu, \sigma^2)$	Probability model for each observation.
μ, σ^2	Unknown population parameters.
n	Sample size.

One equation simultaneously specifies the **observations**, **probability model**, **assumptions**, and **unknown parameters**.

Random Variables, Data and Parameters

For $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$,

Random Variables

$$X_1, \dots, X_n$$

Before sampling:

- Unknown
- Random
- Described by the model

Parameters

$$\mu, \sigma^2$$

- Unknown
- Fixed constants (frequentist)
- Describe the population
- Do not change after sampling

After observing data, $X_i \rightarrow x_i$.

The values that we calculate from the collected x_i 's are merely the **estimates** for the values of the **parameters**.

From One Model to Statistical Models

The previous model

$$X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$$

is only one possible statistical model.

Definition

A statistical model is a family of probability distributions

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\},$$

indexed by an unknown parameter θ .

Interpretation

Each value of θ corresponds to one probability distribution. The data are assumed to arise from one (unknown) member of this family.

Probability Models versus Statistical Models

Probability Model	Statistical Model
Distribution completely specified	Contains unknown parameters
Used to describe randomness	Used to learn from data
Example	Example
$X \sim N(0, 1)$	$X \sim N(\mu, \sigma^2)$
No inference required	Inference required

Key Difference

Statistics begins when the probability model contains **unknown quantities that must be inferred from data**.

Examples of Statistical Models

Blood pressure

$$X_i \sim N(\mu, \sigma^2)$$

Coin tossing

$$X_i \sim \text{Bernoulli}(p)$$

Customer arrivals

$$X_i \sim \text{Poisson}(\lambda)$$

Linear regression

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

Logistic regression

$$P(Y_i = 1) = \frac{e^{\beta^T x_i}}{1 + e^{\beta^T x_i}}$$

Although the mathematical forms differ, every statistical model consists of probability assumptions indexed by unknown parameters.

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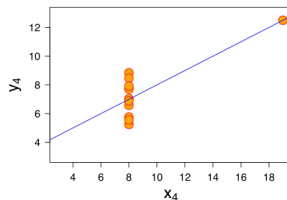
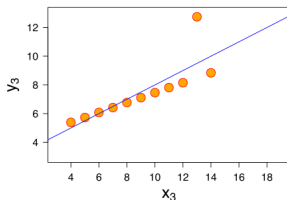
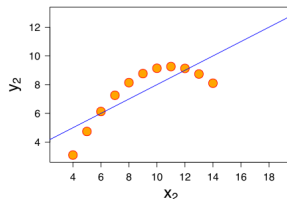
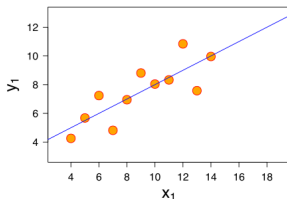
Property	Value	Accuracy
Mean of x	9	exact
Sample variance of x : s_x^2	11	exact
Mean of y	7.50	to 2 decimal places
Sample variance of y : s_y^2	4.125	± 0.003
Correlation between x and y	0.816	to 3 decimal places
Linear regression line	$y = 3.00 + 0.500x$	to 2 and 3 decimal places, respectively
Coefficient of determination of the linear regression: R^2	0.67	to 2 decimal places

All four sets have almost identical statistical parameters. Should we use the same statistical model for these four datasets? **Why?**

Source: Wikipedia

Anscombe's Quartet

The graphs show them to be considerably different. The same model would almost certainly be problematic.



Lessons from Anscombe's Quartet

Although the four datasets have nearly identical summary statistics,

$$\bar{x}, \quad \bar{y}, \quad s_x^2, \quad s_y^2, \quad r, \quad \hat{y} = \beta_0 + \beta_1 x,$$

their structures are fundamentally different.

Statistical Thinking

- Numerical summaries never tell the whole story.
- Every model rests on assumptions.
- Model choice should be guided by both data and scientific understanding.

What Do We Do with a Statistical Model?

Suppose we wish to learn about the average height of all (IITM) campus residents, $\mu = \text{Average height of the population}$. How should we go about it?

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Based on a sample, which should we report?

- A guess (e.g., 5'3")
- Sample mean
- Sample median/mode
- Confidence interval

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Which estimator should we use? How does sample size matter? What makes one estimator better than another? How should we test these hypotheses?

Classes of Statistical Models

Recall that a statistical model is a family of probability distributions

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\},$$

where θ denotes the unknown quantity (or quantities).

Parametric Model

A statistical model is **parametric** if the **unknown parameter is finite-dimensional**, $\theta \in \mathbb{R}^p$, where p is fixed.

Example: Blood Pressure

Assume $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, where $\theta = (\mu, \sigma^2)$.

Only two unknown parameters describe the entire population.

Beyond Parametric Models

Semiparametric Model

Some components are finite-dimensional, while others are left unspecified.

$$X_i \sim F, \quad E(X_i) = \mu,$$

where μ is the parameter of interest but the distribution F is otherwise unknown.

Nonparametric Model

No finite-dimensional parametric form is assumed.

The entire distribution is unknown:

$$X_i \sim F,$$

where F is arbitrary.

Generative Models

A **generative model** specifies the joint distribution

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What can a generative model do?

- Generate new observations.
- Compute conditional probabilities,
- Simulate from the underlying data-generating process.

Examples: Gaussian Mixture Models, Large Language Models (LLMs)

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Why are LLMs "Generative"?

An LLM models $P(w_t | w_1, \dots, w_{t-1})$, the probability of the next token, allowing it to generate entirely new text.

Discriminative Models

A **discriminative model** directly models

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What can a discriminative model do?

- Predict labels or responses.
- Learn decision boundaries.
- Focus on predictive accuracy rather than modelling the data-generating process.

Examples: Logistic Regression, Support Vector Machines, Random Forests, Deep Neural Networks

Key Difference: Marginal distributions assuming X to be given vs. a joint distribution or conditional distribution.

Latent Variable Models

Not every variable in a statistical model is observable. Many models introduce hidden (latent) variables to explain the dependence among the observed variables.

Observed	Latent Variable
Exam scores	

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Observed	Latent Variable
Exam scores	Student ability
Words	Underlying topic
Stock prices	Market state
Gene expression	Biological pathway
Pixel intensities	Object identity

From Models to Data

A statistical model specifies a family of distributions

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}.$$

The data

$$x_1, \dots, x_n$$

have now been observed.

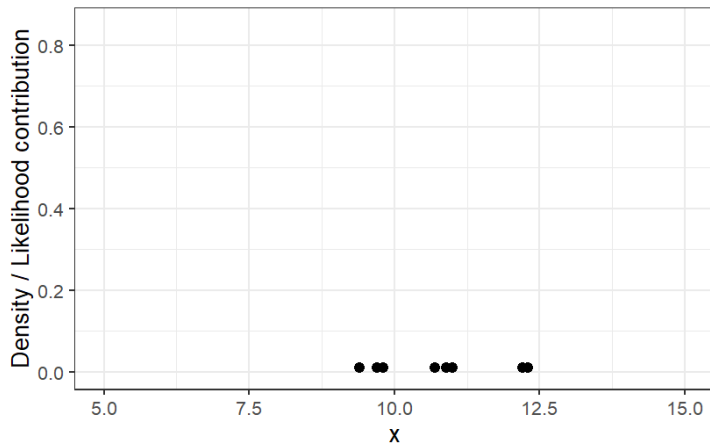
The next question is

Which parameter values are most plausible given the observed data?

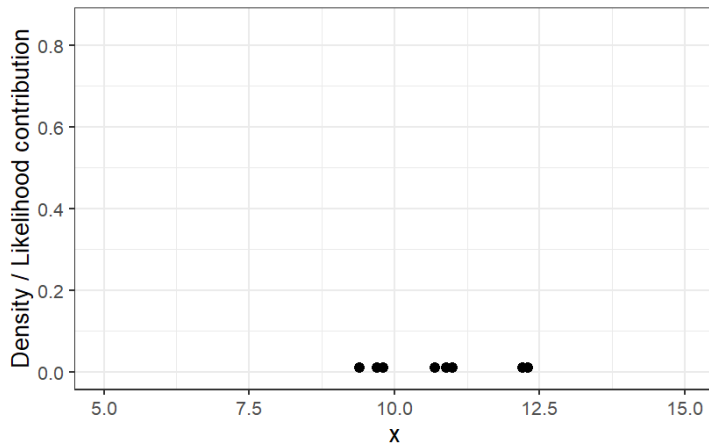
The answer is provided by the

Likelihood Function.

Motivating the next question via an example

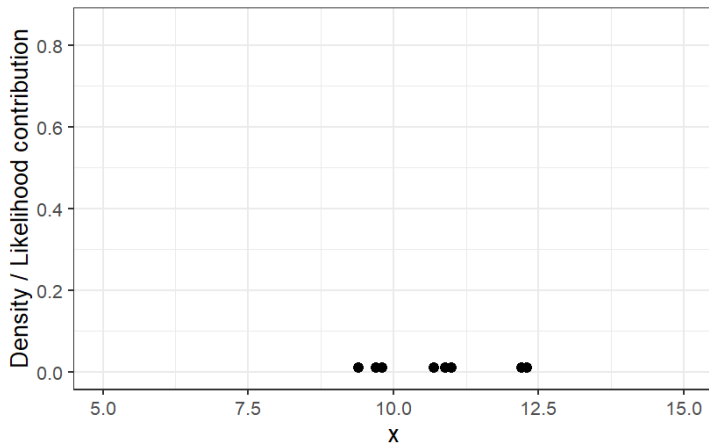


Motivating the next question via an example



Which probability model generated this data?

Motivating the next question via an example



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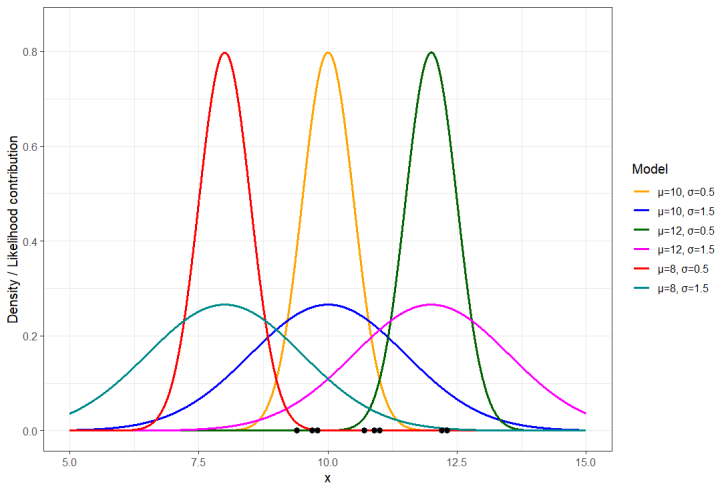
Assume that $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, now what can you tell me?

Motivating the next question via an example (contd.)

Which parameter values are most strongly supported by the observed data?

Motivating the next question via an example (contd.)

Which parameter values are most strongly supported by the observed data?



We will discuss parameter estimation in the next module.